An Organizational Fruit Fly: r/place as a Model Organism for Cumulative Grounded Theory

ABSTRACT

Our field generates claims faster than it can relate them. Biology addresses this through model organisms: the fruit fly, the bread mold, the mouse. Shared data and tools let researchers relate new findings to knowledge of the organism. Our field faces limits that Machlup identified in his AEA Presidential address: cost, observability, and composition in the study of real firms. Organization Science's inaugural editorial described a similar tension between detailed case knowledge and generalizability. We propose a model organism for organizational emergence and evolution: r/place, a 2017 Reddit experiment in which online communities organized on a shared canvas. The dataset is public record: 16.6 million pixel changes and 7.5 million posts and comments across 2,796 subreddits. The instrument is a harness in which one coding agent per community reads that community's full record and codes its organizing. An aggregator composes the coded cases into a population-level qualitative model of 705 curated communities (2.1 million posts and comments and 90% of all pixel changes). The model returns to the coders, who identify where it fails their cases. The aggregator revises the model and identifies organizational forms by grouping communities that request similar changes. Our model organism addresses all three constraints: LLMs reduce the cost of reading the record, researchers can inspect the discussions and pixel changes, and the instrument composes the coded cases while retaining their detail. We report the population model, how we check it, and the remaining risks. Later claims are judged by what they add to this baseline.

Keywords: Knowledge Accumulation; Model Organism; Qualitative Research; Organizational Emergence; Organizational Forms

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Our field struggles to build cumulative knowledge (Pfeffer, 1993; Davis, 2010). Part of the problem is ontological. As March said, "God gave all the easy problems to the physicists" (Dong et al., 2017:12), meaning that organizations are complex, change while we study them, rarely allow experiments, and confound skill with chance. Another part comes from our practices. Our promotion systems institutionalize a "general call to publish or perish" (Aguinis et al., 2020:136). Our journals urge us to "examine outliers (Lewin [1992]) rather than central tendencies" (Daft & Lewin, 1990:6). Jointly, ontology and practice promote the "search for novelty rather than truth" (Barley, 2016:5), leading our field "toward an equilibrium of more rather than better research" (Gartenberg et al., 2026a:795).

Hence, our field produces claims faster than it relates them: a problem of "disorganization, not amount" (Cronin et al., 2021:668). Biology faced the same problem when genome sequencing in the early 2000s produced thousands of claims about gene function (Adams et al., 2000). Each claim met a filter: does it hold in a model organism? Databases grew around each organism (Leonelli, 2016), and researchers judged new claims against everything known about it. "As genetics needs its model organisms, its Drosophila and Neurospora, so [Organization Science] needs standard task environments around which knowledge and understanding can cumulate" (Simon & Chase, 1973:394). Yet we lack a model organism. A model organism is "standardized to fit an integrative and comparative mode of research" (Ankeny & Leonelli, 2011:313): standardized specimens, shared data and tools, and a community working toward an integrated account of the organism. The aim is to understand how its processes connect and which findings extend to other organisms.

We propose one such model organism, a fruit fly for accumulating knowledge on organizational emergence and evolution. We choose this phenomenon because it is where our failure to accumulate shows most. Although "the average organization does not exist", and "significant insights can arise from studying the best or the worst of a population" (Daft & Lewin, 1990:6), lessons drawn from the extremes alone

misdescribe the population they come from (Denrell, 2003). Explaining variation in organizational forms requires relating findings across cases to build "a deeper understanding of how organizational forms come to be different and remain different through time" (Lewin & Volberda, 1999:529), to understand how "organization form[s] emerge instead of being specified in advance" (Daft & Lewin, 1993:iv).

Our model organism is r/place, a 2017 Reddit experiment in which online communities organized on a shared 1,000 by 1,000 pixel canvas (Simpson et al., 2017). It has two parts: a dataset and an instrument. The dataset joins pixel changes from the 72-hour event with public discussions from the event and the following day (Heitmann, n.d.; Reddit, 2017). Each user could change one pixel's color every five minutes and anyone could overwrite it. Communities agreed on designs, divided the work and defended their artworks. Over a million users made 16.6 million pixel changes. The discussion record contains 7.5 million posts and comments across 2,796 subreddits; the Place Atlas records 1,473 artworks (Place Atlas, 2017).

The instrument is a harness in which LLM agents read the record under explicit virtual context management (Packer et al., 2024). One coding agent per community reads that community's full record and codes its organizing; an aggregator composes the coded cases into a population-level qualitative model (Gioia et al., 2013; Grodal et al., 2021). The model returns to the coders, who identify where it fails their cases. The aggregator revises the model and identifies organizational forms by grouping communities that request similar changes. An agent need not hold everything at once: it keeps notes outside its context window, fetches the material a question needs, and no case is dropped to make room for another. Because one agent studies each community, the model can build a census of this curated population[1] (Hannan & Freeman, 1977) while retaining the nuance needed to distinguish how organizational forms differ (McKelvey, 1975).

[1] Each case is one subreddit, a forum on Reddit with its own members and moderators, together with the artworks it built and the comments its members wrote during the event and the day after. Curators documented 777 subreddits that stood behind 1,236 of the 1,473 artworks (Place Atlas, 2017); 705 show evidence of coordination (criterion in the Technical Appendix) and account for 90% of all pixel changes and wrote 2.1 million posts and comments. We quote by comment ID and do not redistribute the raw corpus. Ethics review is underway.

The dataset is frozen. But the instrument can change: the model can improve by changing the literature, broadening the census, or checking how novel mechanisms increase our understanding of r/place. This model organism is an "open-ended program of research" (Lewin & Volberda, 1999:529) in which later models are judged against the baseline, including where they revise or reject it. The call for this special issue asks "what kinds of knowledge can we create when the tools of inquiry themselves change?" (Gartenberg et al., 2026b:2). We address this question by composing case accounts into a population model that preserves their differences.

Our field inherited a limit from economics: studies of single cases do not add up. When interviews and surveys showed that managers do not set prices and wages at the margin (Hall & Hitch, 1939; Lester, 1946), economists replied that "to be important ... a hypothesis must be descriptively false in its assumptions" (Friedman, 1953:14; Machlup, 1946). Machlup (1967) later gave three reasons not to study real firms: cost, observability and composition, since he "might not be able to rely on the composite results obtained from a firm-by-firm analysis" (Machlup, 1967:30). Cyert and March (1963:128) had already answered the first two: their computer model of one department store's pricing, built from interviews with its buyers, made the mechanisms observable and cost only machine time. Composition is the constraint he kept: such a model "has usually not been transferable to other cases" (Machlup, 1967:16). Organization Science's inaugural editorial accepted the same limit, setting "in-depth understanding of a few cases" apart from "generalizability" (Daft & Lewin, 1990:6). Eisenhardt (1989:545) identified the practical obstacle: "With more than 10 cases, it quickly becomes difficult to cope with the complexity and volume of the data."

We focus on qualitative research for three reasons. First, the limit binds hardest there. Qualitative findings rest on case detail that existing approaches struggle to preserve at population scale. Our field remains "largely a science of verbal reports" (Daft & Lewin, 1990:3), and LLMs can now read such reports at population scale. Second, forms that emerge instead of being specified in advance need categories that also emerge from the record. Third, accumulation is least settled there. For some, transparency is a required step toward "replicable and cumulative knowledge" (Aguinis & Solarino, 2019:1292), and knowledge "accumulates vertically, building higher and higher on tested foundations" (Pratt et al., 2020:4); for others,

it accumulates horizontally, as "a stock of broad principles generated across many past situations, each of them unique" (Pratt et al., 2020:5). Costly methods and unobservable phenomena deny others the chance to replicate or extend a finding; results that cannot be related are harder to build on. Our model organism addresses all three constraints: LLMs reduce the cost of reading the record, researchers can inspect the discussions and pixel changes, and the instrument composes the coded cases while retaining their detail.

Humans and LLMs are differently bounded. Humans trade detail about any one case for being able to compose knowledge of many cases (Ragin, 1987). When LLMs form concepts, they compress towards statistical efficiency and drop fine-grained distinctions that humans keep (Shani et al., 2026). They are predisposed "to identify, replicate, and reinforce dominant language and patterns" (Jowsey et al., 2025:1) and are "wildly conservative (tied to past data) as they seek to reduce surprise" (Felin & Holweg, 2024:351). Abduction "starts with the identification of an anomaly, that is, an observation which cannot be explained by existing theories" (Goldenstein et al., 2026:10). Our instrument makes departures from the population model explicit by separating case coding from population composition. Each coder first reads only its own community's full record. When the population model returns, the coder checks it against that record and reports where it needs to change.

This motivates a division of learning. Qualitative researchers aim "to generate novel theoretical insights" (Lê & Schmid, in Gioia et al., 2022:243) and "formulate the issues and troubles that we notice in the world as problems of social science" (Bechky & Davis, 2025:11). LLMs appear "(currently) unqualified to assess whether a conjecture represents a theoretical contribution" (O'Sullivan & Ravasi, 2026:17). We give the community coders a common procedure to follow (Grodal et al., 2021; Than et al., 2025), "so that results are comparable and knowledge can be accumulated" (Günther et al., 2026:863). We judge whether these accounts explain the evidence and contribute to theory. We remain responsible for interpretation and reflexivity.

Our instrument[2] lets the community coders challenge the population model with their case evidence. An initial iteration of the population model stated that communities keep their work by repairing changed pixels. The coder of r/PlaceNL (a community founded to coordinate the Dutch artworks) objected with its community's record: members agreed, "we are going to make blue canal houses and a pretty little windmill on our flag" (r/PlaceNL, post-ID: 636e4q, time: h+68:20, our translation). Yet the community's own defense script kept erasing them: "who, oh who, will turn off the defense script? The little houses are all being wiped away" (r/PlaceNL, comment-ID: dfrvgvk, time: h+71:40, our translation). The coder asked for a revision. The aggregator found similar evidence in 67 communities and, with it, revised the model: repair restores the form the keepers know, so a planned change is erased as damage until the keepers learn of it.

The repository (https://os-fruit-fly.github.io/rplace) holds our instrument, download scripts, every community's coded record and selected aggregator and community checkpoints. Readers can resume an agent from a saved stage and question its work. We call this a checkpoint interview (Technical Appendix). The agent answers with the tools and memory it held then, letting readers compare its explanation with its recorded work rather than trust its "artificial certainty" (Leonardi & Leavell, 2026:516), or accept a black-boxed analysis (Anthony, 2021). The snapshots make the inference inspectable (Aguinis & Solarino, 2019) without rerunning the analysis (Pratt et al., 2020).

The current model's insights (Technical Appendix) are not settled. The agent may write accounts that read well and misread the data. We mitigate this risk at three levels. At the coding level, a script checks code quotations against the source record, and 298 communities were coded twice, once with Opus 5 and once with GPT 5.6-Sol. At the composition level, human audits can test selected cross-case inferences. We

[2] AI use. AI agents carry out the analysis. Community coders and their fleets are OpenAI's GPT 5.6-Sol, run in Codex; the coders moved to GPT 6.1-Sol for the return to the model and, after a naming step, for their communities' local models. A separate agent on GPT 6.1-Sol read the r/place subreddit, and its reading added two elements and an arrow to the model. The aggregator is Anthropic's Claude Opus 5.5 in Claude Code, with DeepSeek's V4.1 Flash for routine reading checks and its fleet of readers. The authors designed the instrument and ruled on questions during the run. The repository records model identifiers, settings, prompts, interventions and logs. Claude and Codex also helped edit the text and build the figures and website.

will also build two further population models from the same record: an algorithm-supported induction that clusters communities on term frequencies, with no coder-in-the-loop (Shrestha et al., 2021); and a second aggregator that composes from cross-case tables (Eisenhardt, 1989). At the source level, we omitted the r/place subreddit from the population model: it was the venue where communities met and traded, not one of the communities building on the canvas. It holds more posts than any other subreddit, and many record what different communities were doing. We will finish coding it and use it as a witness to the event, to triangulate the model against an independent record (Eisenhardt, 1989; Gioia et al., 2013).

Two risks remain. Checkpoint interviews depend on continued access to compatible models and software. We will therefore rerun the full pipeline with an open-weight model and archive its weights. Cost is the second: coding the 705 communities plus one composition run took around three months with a US$200 subscription; a census of 2,796 would cost a year, so we kept the curated list. If costs fall with later model generations, future runs could broaden the census or compare improved versions of the population model.

As with biology's understanding of the fruit fly, our understanding of what happened during r/place 2017 will grow over time. A model organism does not forbid outliers or novelty. It gives them a baseline to be measured against. Machlup's constraints were an economist's excuse to disregard the behavior of organizations. Organization Science has no such excuse: it "was founded as a bet on escaping the constraints of local search and incrementalism" (Gartenberg et al., 2026b:1). Whether our field "evolves toward 'more' or 'better'" is an institutional question (Gartenberg et al., 2026a:811), and a model organism is a nudge toward "better".

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